Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery - Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

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Project Proposal Report

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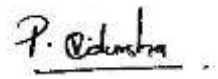
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Healthcare documentation plays a vital role in ensuring patient safety, treatment continuity, and effective communication among caregivers. One important part of this process is the recording of Activities of Daily Living (ADL), such as medication administration, food and fluid intake, mobility assistance, hygiene support, and emotional observations. However, in many healthcare environments, caregivers still depend on manual documentation methods, which are time-consuming, burdensome, and sometimes incomplete. This reduces the time available for direct patient care and may also affect the quality of shift handover communication.

This research proposes a Context-Aware Intelligent System for Automated Extraction, Structuring, and Real-Time Documentation of Activities of Daily Living (ADL) from Caregiver Voice Logs. The proposed system is designed to convert caregiver speech into structured ADL records automatically. It integrates several intelligent components, including speech recognition, auto-editing for handling fillers and spoken corrections, domain-specific natural language processing for ADL information extraction, and context-aware analysis for improved interpretation. In addition, the system generates structured care reports and shift handover summaries in text, speech, and 3D avatar video form to support continuity of care.

The study follows a Design Science Research approach with experimental evaluation. The system will be developed using AI-based speech and language technologies and evaluated using performance metrics such as Word Error Rate, precision, recall, and F1-score, together with quality measures for generated reports and handover outputs. Ethical considerations such as privacy, confidentiality, and secure handling of caregiver voice data will also be addressed.

The expected outcome of this research is a practical and intelligent documentation system that can reduce manual workload, improve the accuracy and completeness of ADL records, and support better communication between caregivers. The study is aligned with SDG 3: Good Health and Well-Being and contributes to the Artificial Intelligence and Data Science research cluster by combining speech processing, NLP, context-aware computing, and intelligent healthcare reporting into one integrated solution.

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We are especially grateful to the Lanka Professional Caregivers Association for allowing us to observe caregiver training practices and gain a deeper understanding of the daily responsibilities and challenges faced by caregivers in healthcare environments. These insights were highly valuable in helping us design a more practical and user-centered solution.

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TABLE OF CONTENT

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
TABLE OF CONTENT 5
LIST OF TABLES 7
LIST OF FIGURES 8
LIST OF APPENDICES 9
LIST OF ABBREVIATIONS 10
1. INTRODUCTION 11
1.1 Background and Context 11
1.2 Literature Review 13
1.3 Research Gap 15
2. OBJECTIVE 17
2.1 Main Objective 17
2.2 Specific Objectives 17
3. METHODOLOGY 18
3.1 Research Approach 18
3.2 Technical Methods and Algorithms 18
3.3 Tools and Platforms 19
3.4 Data Collection and Ethical Considerations 19
3.5 Validation Strategy and Performance Metrics 20
4. HIGH-LEVEL SYSTEM ARCHITECTURE 21
5. USER REQUIREMENTS 22
5.1 Stakeholder Identification 22
5.2 Functional Requirements 22
5.3 Non-Functional Requirements 23
6. COMMERCIALIZATION PLAN 24
6.1 Target Market and Customer Persona 24
6.2 Value Proposition 24
6.3 Revenue Model 24
6.4 Cost Estimation 25

6.5 Pricing Strategy 25
6.6 Competitive Advantage 25
6.7 Intellectual Property Considerations 26
7. BUDGET AND JUSTIFICATION 27
7.1 Hardware Costs 27
7.2 Software Licenses 27
7.3 Cloud Services 27
7.4 Human Effort Cost 28
7.5 Estimated Budget Summary 28
8. WORK BREAKDOWN STRUCTURE 29
9. GANTT CHART 30
10. AI USAGE DISCLOSURE 31
11. REFERENCES LIST 32
12. APPENDICES 33

LIST OF TABLES

Table 1 List of Abbreviations 10

LIST OF FIGURES

Figure 1 Overview of the integrated AI-based caregiving system showing data flow and component interactions. 21
Figure 2 High-Level System Architecture for My Component 21
Figure 3 Work Breakdown Structure 29
Figure 4 Gantt Chart 30

LIST OF APPENDICES

Appendix 1 Difficulties Faced by Caregivers When Recording Daily Patient Care Activities 33
Appendix 2 Caregivers' Views on the Usefulness of a Voice-Based Documentation System.....33
Appendix 3 Frequency of Documentation Delays Due to Workload.....33

LIST OF ABBREVIATIONS

Abbreviation	Description
ADL	Activities of Daily Living
AI	Artificial Intelligence
ASR	Automatic Speech Recognition
NLP	Natural Language Processing
NER	Named Entity Recognition
TTS	Text-to-Speech
EHR	Electronic Health Record
ML	Machine Learning
DL	Deep Learning

Table 1 List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Healthcare documentation is a very important part of patient care because it records what care was provided, what changes were observed, and what actions need to be followed later. In caregiving settings such as hospitals, nursing homes, and home-care environments, caregivers often need to document Activities of Daily Living (ADL), such as eating, drinking, medication intake, mobility, hygiene, sleep, and emotional condition. These records are important because they help healthcare staff monitor patient progress, support treatment decisions, and maintain continuity of care between different caregivers [1], [3]. Clinical documentation research also shows that both structured documentation and narrative documentation are needed in healthcare, since structured records support research and decision systems, while narrative records remain essential for communication and post-hospital care [2].

The real-world problem is that documentation takes a large amount of time and often interrupts direct caregiving work. One study reports that nurses spend around 25% to 41% of their working hours on documentation, showing that documentation is a major burden in care environments [1]. Another study reports that physicians spend about 44% of their daily working hours on electronic health record management, with a significant amount of that time spent on documentation tasks [2]. This issue is important because it affects caregivers through stress, workload, and workflow interruption, and it affects patients because delayed or incomplete records can reduce care quality and increase the chance of communication errors [1], [2].

The problem becomes even more serious during handover between shifts. Accurate and timely records are necessary for the next caregiver to understand the patient's recent condition, completed care activities, and any new concerns. Research on discharge documentation explains that complete and accurate summaries are essential for continuity of care, and delayed documentation may increase the risk of medication errors and rehospitalization [2]. Although discharge summaries are not the same as ADL logs, the same idea applies in daily caregiving, if information is missing or unclear, the next caregiver may not fully understand the patient's current situation [2].

Existing solutions have improved documentation to some extent, but they still have important limitations. Some systems support speech-based documentation, and some recent studies use large language models to generate summaries from conversations or structured clinical data [1], [2]. However, these solutions are mainly designed for transcription or summary drafting. They usually do not fully understand the context of caregiver speech, they do not structure the information into ADL categories automatically, and they do not handle spoken corrections, fillers, or hesitations in a reliable way [1], [2]. In addition, older rule-based approaches for clinical text generation were difficult to scale across medical domains, while newer LLM-based approaches still depend heavily on data quality and often require human revision [2].

Another major challenge is speech recognition in healthcare environments. Real caregiving speech often includes accent variation, natural hesitation, repeated words, spoken repairs, background noise, and privacy-sensitive content. A recent ASR study found that even a strong multilingual speech model degraded to 40.94% word error rate when tested on real clinical

telephony speech, making it unsuitable without adaptation [5]. The same study reported a 17.1% relative improvement after adaptation and also emphasized that acoustic adaptation is essential and cannot be replaced by language models alone [5]. This shows that ordinary ASR is not enough for safe and dependable healthcare documentation [5], [6].

Research on ADL extraction also shows promise, but there are still clear limitations. A systematic review identified 22 studies that used NLP to extract ADL information from unstructured electronic health records. The review found that NLP is promising, but it also reported challenges such as restricted access to EHR data, inadequate documentation, lack of granularity, and small datasets [3]. This means that current work supports ADL extraction from written clinical notes, but not fully from direct caregiver voice logs [3].

To address these limitations, this research proposes a Context-Aware Intelligent System for Automated Extraction, Structuring, and Real-Time Documentation of ADL from Caregiver Voice Logs. The proposed system will first use ASR to convert caregiver speech into text. Then it will apply disfluency and correction handling to remove fillers, repetitions, and spoken selfcorrections. After that, a domain-specific NLP and named entity recognition module will extract important ADL details such as medication, food intake, mobility activity, symptoms, time, and emotional observations. A context-aware module will then improve interpretation by considering patient history, time of day, and care-related context. Finally, the system will generate structured ADL records and handover-ready summaries for caregivers [3],[5].

The context-aware component is especially important because medical information often cannot be interpreted correctly without surrounding context. A scoping review of context-aware medical systems found that the most common useful contexts in healthcare are user location, demographic information, movement or activity level, time of day, lab or vital data, and patient history [4]. The same review explains that medical data may appear normal or dangerous depending on the situation, which means that context is necessary for meaningful interpretation [4]. This idea is highly relevant to ADL voice documentation, because the same spoken event may need different interpretation depending on the patient's status and care situation [4].

This research aligns with Sustainable Development Goal 3: Good Health and Well-Being, because it aims to improve care quality, reduce the documentation burden on caregivers, and support safer communication between healthcare workers. It also aligns with the Artificial Intelligence and Data Science research cluster because it combines ASR, NLP, disfluency detection, context-aware computing, and intelligent healthcare reporting into one applied AI system [3],[5].

1.2 Literature Review

Recent research shows growing interest in reducing healthcare documentation burden through AI, speech technology, and natural language processing. One important study proposed a care documentation support system that uses audio recordings of care actions and large language models to generate summaries from nurse-patient dialogues [1]. This study is highly relevant because it focuses on care actions rather than only doctor-patient consultations. It also highlights that documentation takes a large share of caregivers' working time and shows that nurse-patient dialogues contain important information about performed care actions, physical condition, and mental state [1]. This supports the idea that speech-based care documentation is both useful and necessary [1].

Another important area is the automatic generation of clinical summaries. A recent study on discharge summary generation used LLaMA3 and structured clinical data to produce editable draft summaries for physicians [2]. The results showed that AI-generated summaries can be helpful, but the study also found an average of 2.84 mistakes per summary, and important information such as patient history, lifestyle factors, and intraoperative findings was often missing [2]. The paper further reported that only 54% of the content in physician-written summaries was available in the structured dataset used for generation, showing that summary quality depends heavily on the completeness of the input data [2]. This is important for the present study because it suggests that reporting tools can be useful only when the system first captures the right information correctly [2].

For ADL-related information extraction, a systematic review found 22 studies that used NLP to extract ADL information from unstructured EHR notes [3]. Most of these systems used deep learning together with rule-based or machine learning methods, and most studies reported relatively high performance scores [3]. However, the review also explains that comparing systems is difficult because datasets, ADL categories, preprocessing methods, and evaluation approaches vary widely. It also highlights common limitations such as inadequate documentation, restricted access to EHR data, lack of granularity, and small datasets [3]. This shows that although NLP is promising for ADL extraction, the current literature mainly addresses written clinical notes rather than spontaneous caregiver speech [3].

Context-aware healthcare research also provides strong support for the proposed work. A systematic scoping review identified 25 context-aware medical systems that had been used by healthcare providers or patients [4]. The review found that the most common useful contexts were user location, patient history, movement or activity level, time of day, demographic information, and lab or vital data [4]. It also explains that medical data often cannot be interpreted correctly without situational context. For example, the same heart rate may be safe during exercise but dangerous during sleep [4]. This is important for the present research because caregiver speech logs also need context to be interpreted correctly and transformed into meaningful ADL documentation [4].

Speech recognition research further shows that healthcare environments present serious technical challenges. A recent clinical telephony ASR study found a large "reality gap" between benchmark performance and real-world healthcare audio, with a strong multilingual model

reaching 40.94% WER on rural clinical speech [5]. The same paper explains that ASR errors are especially risky in healthcare because they may affect medically important information such as dosage, negation, and symptom reporting [5]. It also reports that parameter-efficient adaptation improved performance by 17.1% and reduced catastrophic forgetting by 55% [5].

This literature supports the need for domain-adapted ASR in healthcare, but it still focuses mainly on transcription accuracy rather than full ADL documentation [5].

Overall, the literature shows clear progress in speech-based healthcare documentation, summary generation, ADL extraction, context-aware healthcare systems, and ASR adaptation. However, these research areas are still mostly separate. Some studies focus on summarization, some focus on written ADL extraction, and others focus on speech recognition performance. Very few studies combine all of these parts into one end-to-end system designed specifically for caregiver voice logs [1],[5].

1.3 Research Gap

Although previous studies have shown progress in speech recognition, clinical summarization, ADL extraction, and context-aware healthcare systems, there is still a clear research gap in the area of real-time caregiver voice documentation. Most existing ASR studies mainly focus on improving transcription accuracy in difficult healthcare environments [5], [6]. This is useful, but transcription alone is not enough for caregiving documentation. A raw transcript only shows what was spoken; it does not automatically identify the actual care activity, organize the information into ADL categories, or produce structured records that can be used directly for reporting and shift handover [5]. In real caregiving situations, spoken language is often informal and may contain hesitations, repeated words, false starts, and self-corrections. Therefore, a major missing component in current systems is auto-editing, where the system can detect spoken corrections such as "no," "sorry," or "actually," remove the incorrect earlier phrase, and keep only the caregiver's final intended meaning [7], [8].

Another clear gap is found in the ADL extraction literature. Existing NLP studies show that ADL-related information can be extracted from unstructured electronic health records, but most of these studies begin with written clinical text rather than direct caregiver speech [3]. This means that an important step is still missing between spoken caregiver input and structured ADL documentation output. Caregiver voice logs are more difficult than written notes because they are less structured, more spontaneous, and often contain disfluencies. As a result, current ADL extraction methods do not fully solve the problem of converting natural caregiver speech into meaningful, structured care records [3], [7].

There is also a gap in current automatic documentation systems. Existing systems often focus on doctor-patient conversations, discharge summaries, or AI-supported dictation workflows [1], [2], [10]. These approaches are valuable, but they do not fully reflect the daily caregiving environment, where caregivers record observations while performing routine care activities such as feeding, giving medication, helping with hygiene, or assisting movement. In addition, recent summary generation systems still depend strongly on the completeness of the source data and may produce missing details or factual inaccuracies when the input is not rich enough [2]. This shows that current solutions are still insufficient for caregiver-centered ADL documentation [1], [2].

Context-aware healthcare research also reveals another limitation. Although context-aware systems are increasingly being used in healthcare, many of them are designed for monitoring, reminders, rehabilitation, navigation, or decision support rather than for care documentation [4]. Very few systems use context specifically to understand caregiver voice logs and improve ADL interpretation using factors such as patient history, time of day, activity type, or previous care events [4]. Without this contextual understanding, even a correctly transcribed sentence may still be incomplete or unclear from a care-documentation perspective.

Therefore, the main research gap is the lack of an integrated caregiver-centered system that can capture caregiver speech, transcribe it, automatically edit spoken corrections and disfluencies, extract ADL-related information, use context to improve interpretation, and then generate structured ADL records together with handover-ready summaries. The novelty of this research is not simply the use of ASR or NLP as separate tools. Instead, the novelty lies in combining

speech recognition, auto-editing, domain-specific ADL extraction, context-aware understanding, and intelligent reporting into one unified framework designed specifically for caregiving environments [1],[8].

In summary, current approaches are mostly transcription-focused, text-focused, or limited to only one stage of the documentation workflow. They do not fully transform natural caregiver speech into clear, structured, and context-aware ADL documentation. This study addresses that missing area by proposing a system that goes beyond speech-to-text and supports the full process from caregiver voice input to meaningful care reporting and shift handover [1],[8].

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a context-aware intelligent system that can automatically convert caregiver voice logs into structured Activities of Daily Living (ADL) documentation, with the purpose of improving documentation accuracy, reducing manual workload, and supporting more effective care reporting and shift handover in caregiving environments.

2.2 Specific Objectives

1. To examine the current methods used by caregivers to record ADL information and to identify the main challenges associated with manual documentation, including delayed recording, incomplete notes, inconsistency, and increased administrative burden.

2. To develop a speech recognition component that can accurately convert caregiver voice logs into text by recognizing care-related vocabulary, routine caregiving expressions, and naturally spoken language commonly used in healthcare and caregiving settings.

3. To design and implement an auto-editing component that can identify and handle fillers, hesitations, repetitions, and spoken self-corrections in caregiver speech, so that only the final intended meaning is preserved for clear and reliable documentation.

4. To develop a domain-specific natural language processing module that can automatically extract and structure key ADL-related information from caregiver voice logs, including medication details, food and fluid intake, mobility assistance, hygiene activities, symptoms, time references, and emotional observations.

5. To incorporate context-aware analysis into the proposed system in order to improve the interpretation of caregiver voice logs by using relevant contextual information such as patient history, activity type, time-related information, and previous care records.

6. To generate structured ADL records and concise shift handover summaries from the extracted information, and to evaluate the proposed system in terms of accuracy, completeness, usability, and its ability to support continuity and quality of patient care.

3. METHODOLOGY

3.1 Research Approach

This research will follow a Design Science Research approach together with an experimental evaluation method. Design Science is suitable because the main purpose of this study is to design and develop a practical intelligent system that solves a real-world problem in healthcare documentation. In this research, the main artifact is a context-aware system that can convert caregiver voice logs into structured Activities of Daily Living (ADL) records and generate handover-ready care reports. The experimental aspect is important because the developed system must be tested to measure how accurately and effectively it performs in realistic caregiving situations. This approach is appropriate for the proposed study because earlier research has also focused on building and evaluating healthcare documentation systems, ADL extraction methods, and AI-supported reporting tools [1],[4].

3.2 Technical Methods and Algorithms

The proposed system will be developed as a multi-stage intelligent pipeline. First, a speech recognition module will convert caregiver voice logs into text. Since caregiver speech may contain healthcare-related vocabulary, natural spoken expressions, and accent variation, a speech recognition model that supports healthcare-related transcription will be used [5], [6]. Second, an auto-editing module will identify and remove disfluencies such as fillers, hesitations, repetitions, and spoken self-corrections. This is important because caregivers may naturally say phrases such as “no,” “sorry,” or “actually,” and the system must preserve only the final intended meaning for correct documentation [7], [8].

Third, a domain-specific NLP module will process the cleaned text and extract important ADLrelated information. This module will use techniques such as Named Entity Recognition (NER), text classification, and rule-based extraction to identify key details such as medication, dosage, food and fluid intake, mobility support, hygiene activities, symptoms, time references, and emotional observations [3]. Fourth, a context-aware analysis module will improve the interpretation of the extracted information by using relevant contextual data such as patient history, time of day, care activity type, and previous care records [4]. Fifth, an intelligent reporting module will generate structured ADL records and daily care reports from the extracted data [1], [2].

In addition, the system will include a shift handover generation module that creates both textbased and spoken handover summaries. The handover summary text will be converted into speech using a Text-to-Speech (TTS) component. After that, the spoken output will be synchronized with a 3D avatar to generate a short handover video presentation. This will allow the next caregiver to receive a quick and clear overview of the patient’s recent condition and care activities in both text and video form. The 3D avatar-based handover is expected to improve understanding, support continuity of care, and make the handover process more engaging and efficient.

3.3 Tools and Platforms

The proposed system will be developed mainly using Python because it provides strong support for speech processing, natural language processing, machine learning, and system integration. Python is widely used in artificial intelligence research and is suitable for building a prototype that combines multiple intelligent components in one workflow. Google Colab will be used as the main development platform because it is easy to access, convenient for experimentation, and suitable for developing and testing AI-based models in an academic research environment.

For the speech recognition component, the study will use Whisper because it is a well-known speech-to-text model that performs effectively in converting spoken language into text and can support healthcare-related vocabulary through domain adaptation [6]. For the natural language processing component, spaCy, Hugging Face Transformers, and NLTK will be used for text preprocessing, named entity recognition, and domain-specific text analysis. These tools are appropriate because they provide strong support for extracting important ADL-related information from caregiver speech transcripts.

For the Text-to-Speech (TTS) component, the study will use gTTS and pyttsx3 to convert generated handover summaries into spoken output. These tools are useful because they allow the system to produce audio-based handover communication in addition to text-based reports. For the 3D avatar video generation, a suitable avatar platform will be integrated so that the synthesized voice can be synchronized with a visual avatar presentation. This is important because the proposed system does not generate only written outputs, but also aims to provide a short spoken handover summary through a 3D avatar to improve communication between caregivers.

During the prototype stage, the structured outputs generated by the system will be stored in a database so that ADL records, extracted information, and generated summaries can be organized, managed, and evaluated efficiently. Overall, these tools and platforms are selected because they are practical for research, suitable for intelligent system development, and capable of supporting both the text-based and multimedia components of the proposed solution.

3.4 Data Collection and Ethical Considerations

The data for this research will mainly consist of caregiver voice logs collected for the purpose of developing and evaluating the proposed system. These voice logs will contain spoken information related to common Activities of Daily Living (ADL), such as medication administration, food and fluid intake, mobility assistance, hygiene support, symptom observation, sleep-related notes, and emotional condition reporting. The aim is to collect caregiver speech that reflects the type of language normally used during care activities, since the proposed system is designed to process real caregiver voice input and transform it into structured documentation.

The collected voice data will then be transcribed and used for system development and testing. To make the dataset more useful, the recordings should include natural spoken language patterns such as pauses, fillers, hesitations, repetitions, and spoken corrections, because these are common in real caregiving communication. This is important for the proposed research, since one of the system components is the auto-editing module, which is expected to identify spoken corrections and preserve only the final intended meaning. In some cases, simulated

caregiving scenarios may also be used to supplement the dataset if additional examples are needed for testing specific ADL situations, but the main focus of the study is caregiver voice data.

Ethical considerations are very important in this research because caregiver voice logs may include sensitive patient-related information. If voice data is collected from real participants, informed consent must be obtained before recording begins. Participants should be clearly informed about the purpose of the study, the use of the recordings, and the steps taken to protect privacy and confidentiality. Any personally identifiable information must be removed before analysis, and all collected data must be anonymized to ensure that individuals cannot be identified from the dataset.

The data must also be stored securely and used only for research purposes. Since healthcarerelated information is private and sensitive, proper precautions must be taken to prevent unauthorized access, misuse, or disclosure of the recordings and generated records. Previous studies in healthcare AI and clinical speech processing have also highlighted privacy protection and secure handling of sensitive voice data as important ethical requirements [2], [5]. In addition, if institutional ethical approval is required, it must be obtained before data collection begins.

Overall, this study will follow important ethical principles such as informed consent, privacy protection, confidentiality, anonymization, and responsible data use. By collecting caregiver voice logs carefully and handling the data securely, the research can support the responsible development of the proposed context-aware ADL documentation system.

3.5 Validation Strategy and Performance Metrics

The proposed system will be validated by evaluating each main component separately and then assessing the overall quality of the final output. For the speech recognition component, performance will be measured using Word Error Rate (WER) to determine how accurately caregiver speech is converted into text [5]. For the auto-editing component, precision, recall, and F1-score will be used to measure how well the system detects fillers, repetitions, and spoken corrections [7], [8].

For the ADL extraction component, precision, recall, and F1-score will also be used to evaluate how accurately the system identifies important information such as medication, food intake, mobility, hygiene activities, symptoms, time references, and emotional observations [3]. The generated ADL records and handover summaries will then be evaluated based on correctness, completeness, clarity, and usefulness [1], [2].

In addition, the Text-to-Speech output and 3D avatar handover video will be assessed based on speech clarity, understandability, and presentation quality. Overall, this validation strategy is important because it measures both the technical performance of the system and the practical usefulness of the final documentation and handover output.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

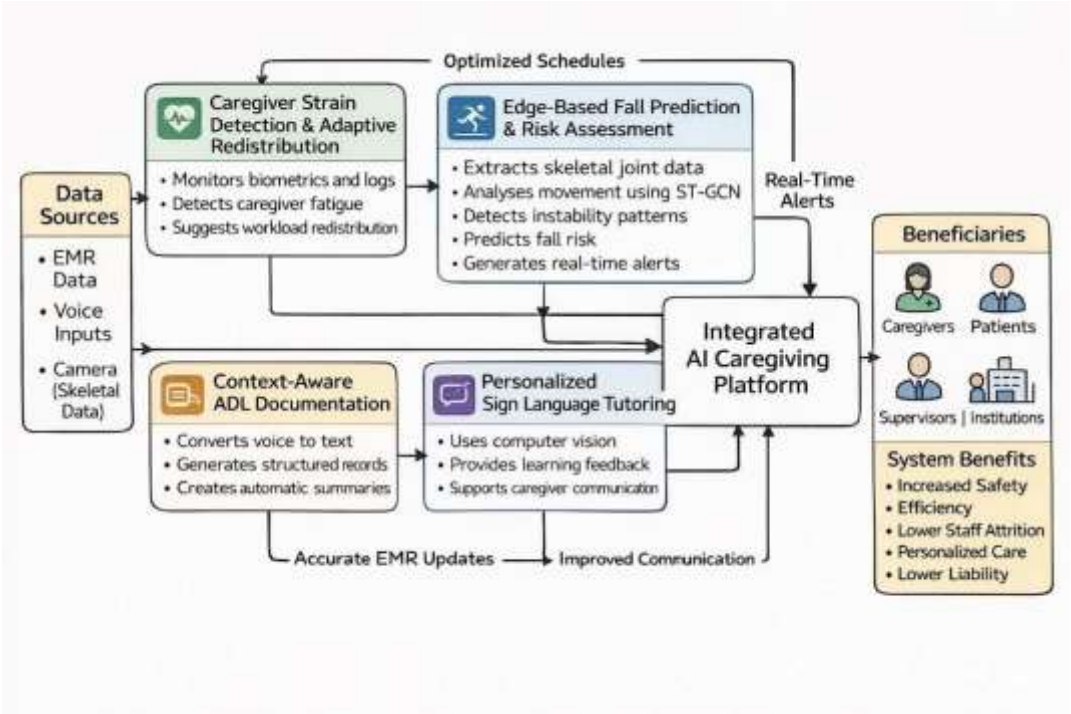


Figure 1 Overview of the integrated AI-based caregiving system showing data flow and component interactions.

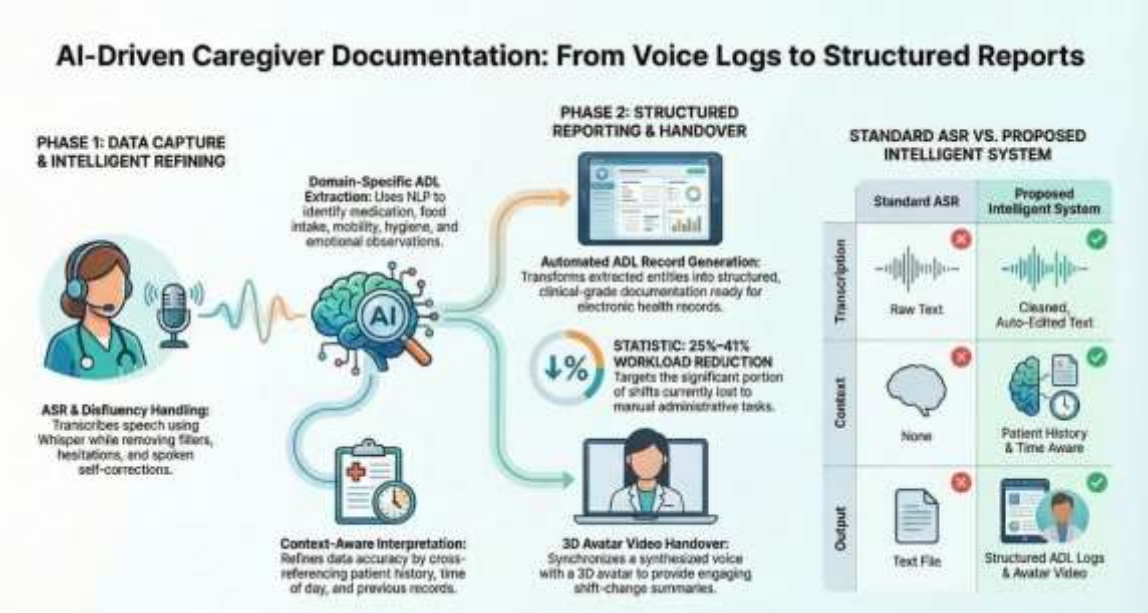


Figure 2 High-Level System Architecture for My Component

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The main stakeholders of the proposed system are caregivers, patients, healthcare administrators and supervisors. Caregivers are the primary users because they will use the system to record voice logs, generate ADL documentation, and access handover information. Patients are indirect stakeholders because the quality and accuracy of the generated records can affect continuity of care and patient safety. Healthcare administrators, supervisors, or nursing managers are also important stakeholders because they may use the generated records and reports for monitoring care quality, reviewing documentation, and supporting workflow improvement. In addition, future system evaluators such as researchers or healthcare experts may also be involved in validating the usefulness and accuracy of the system.

5.2 Functional Requirements

1. The system shall allow caregivers to record voice logs related to daily care activities.
2. The system shall convert caregiver voice input into text using a speech recognition component.
3. The system shall identify and remove fillers, hesitations, repetitions, and spoken self corrections through an auto-editing process.
4. The system shall extract key ADL-related information such as medication, food and fluid intake, mobility, hygiene activities, symptoms, time references, and emotional observations.
5. The system shall organize the extracted information into structured ADL records.
6. The system shall use contextual information such as patient history, care activity type, and time-related details to improve interpretation of caregiver speech.
7. The system shall generate daily care reports from the structured ADL data.
8. The system shall generate shift handover summaries in text form.
9. The system shall convert handover summaries into speech using a Text-to-Speech component.
10. The system shall present the spoken handover summary through a 3D avatar video interface.
11. The system shall store generated ADL records and summaries in a database for retrieval and review.
12. The system shall allow authorized users to access previously generated documentation and handover outputs.

5.3 Non-Functional Requirements

- Security

  The system must protect caregiver voice logs, patient-related information, and generated ADL records from unauthorized access. All stored data should be handled securely to maintain privacy and confidentiality.

- Performance

  The system should process caregiver voice logs within a reasonable time so that the generated ADL records and handover summaries can support near real-time caregiving activities.

- Accuracy

  The system must provide accurate speech recognition, auto-editing, and ADL extraction results because incorrect documentation may affect care quality and communication between caregivers.

- Usability

  The system should be easy to use, with a simple and clear interface, so that caregivers can operate it without needing advanced technical knowledge.

- Reliability

  The system should function consistently and produce stable results during repeated use in caregiving environments.

- Maintainability

  The system should be designed in a way that makes it easier to update, improve, and expand in the future, such as adding new ADL categories or improving the reporting features.

- Scalability

  The system should be able to handle a larger amount of caregiver voice logs and documentation records if the number of users or patients increases.

- Clarity of Output:

  The generated ADL records, text summaries, speech outputs, and 3D avatar handover presentations should be clear, understandable, and useful for caregiving communication.

- Availability

  The system should be available whenever caregivers need to document care activities, especially during busy working hours and shift handover periods.

- Privacy

  The system must ensure that sensitive healthcare-related data is used only for authorized purposes and that personal information is anonymized when required for research or evaluation.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed system is mainly targeted at healthcare organizations that depend on regular patient documentation and smooth communication between caregivers. This includes hospitals, nursing homes, elder-care centers, rehabilitation centers, assisted living facilities, and homecare service providers. These organizations often handle a large number of daily care activities and require accurate ADL documentation to maintain patient safety and continuity of care.

The main customer persona is a healthcare institution that wants to reduce the time spent on manual documentation, improve the quality of care records, and strengthen shift handover communication. Within such organizations, the direct users of the system will be caregivers, nurses, and support staff who record daily care activities. The main decision-makers are likely to be hospital administrators, nursing managers, care home owners, clinical supervisors, and digital health coordinators who are responsible for operational efficiency, compliance, and service quality.

6.2 Value Proposition

The proposed system offers a strong value proposition by transforming caregiver voice logs into structured ADL documentation automatically. This helps reduce the burden of manual charting, saves time for caregivers, and improves the completeness and consistency of care records. Instead of relying only on handwritten notes, later typing, or basic speech-to-text tools, the system provides a more intelligent solution that can recognize caregiving activities, clean spoken corrections, extract important ADL information, and generate organized documentation.

A major value of the system is its support for continuity of care. In addition to structured ADL records, the system can generate handover summaries in text, speech, and 3D avatar video form. This can help incoming caregivers quickly understand the patient's recent condition and completed care activities. As a result, the system does not only improve documentation, but also supports safer communication, better workflow, and more informed caregiving decisions.

6.3 Revenue Model

A Software-as-a-Service model is the most suitable revenue model for this system. Under this model, healthcare organizations would pay a subscription fee to access the platform and its features. The subscription can be offered on a monthly or yearly basis, depending on the size of the organization and the level of service required.

Additional revenue can also come from enterprise licensing for large hospitals or healthcare networks, where the system is deployed across multiple departments or care units. Other possible income sources include setup charges, system customization, staff training, maintenance services, and premium feature access. For example, advanced reporting, predictive analytics, or 3D avatar handover features can be included in higher-value service packages.

6.4 Cost Estimation

The main costs involved in commercializing the system will include software development, model training and testing, interface design, database integration, system maintenance, and user support. Additional costs may arise from Text-to-Speech integration, 3D avatar generation, deployment infrastructure, and security implementation for handling healthcare-related data. There may also be expenses related to data collection, ethical approval, domain expert consultation, and system validation.

At the prototype stage, the costs will mainly be related to research, development, and testing. At the commercialization stage, the costs will expand to include product deployment, cloud or server resources, technical support, updates, user onboarding, and scaling the system for larger healthcare organizations. Therefore, careful cost planning is important to ensure that the system remains both practical and financially sustainable.

6.5 Pricing Strategy

A tiered pricing strategy would be appropriate for the proposed system. This means that different service packages can be offered based on the needs and size of the customer. A basic package can be designed for smaller care centers or home-care providers and may include voice logging, transcription, and structured ADL record generation. A standard package can include context-aware analysis and intelligent reporting features. A premium package can offer advanced features such as predictive insights, spoken handover summaries, and 3D avatar video handover support.

This pricing strategy is beneficial because it allows the system to be affordable for smaller healthcare providers while also creating higher-value plans for larger institutions with more complex documentation needs. It also gives customers the flexibility to choose a package that matches their budget and operational requirements.

6.6 Competitive Advantage

The main competitive advantage of the proposed system is that it is designed specifically for caregiving environments rather than general speech documentation. Many existing tools focus only on speech-to-text conversion or basic dictation support. In contrast, this system goes beyond transcription and provides a complete care documentation workflow. It can capture caregiver voice input, apply auto-editing, extract domain-specific ADL information, use context-aware interpretation, and generate structured documentation together with handoverready summaries.

Another important advantage is the inclusion of multimedia handover support through text, speech, and 3D avatar video. This makes the system more interactive and more useful in practical caregiving settings. Because the system focuses on caregiver workflow, ADL documentation, and continuity of care, it provides a more specialized and innovative solution compared to standard dictation platforms or general healthcare reporting tools.

6.7 Intellectual Property Considerations

The proposed system contains several components that may have intellectual property value. These include the overall system design, the ADL documentation workflow, the auto-editing mechanism for caregiver speech, the context-aware interpretation process, the structured reporting method, and the 3D avatar-based handover generation approach. The software code, models, documentation, and interface design can be protected through copyright.

If the final implementation includes a sufficiently novel technical method, there may also be future potential for patent-related consideration, especially for the integration of auto-editing, context-aware ADL extraction, and avatar-based handover generation in a healthcare documentation system. In addition, the product name, brand identity, and logo can be protected through trademark registration if the system moves toward commercial release. It is also important to ensure proper licensing compliance for any third-party frameworks, pretrained models, Text-to-Speech tools, or avatar technologies used in the system.

Overall, the commercialization potential of this system is strong because it addresses a real and costly problem in healthcare. By reducing documentation burden, improving care communication, and supporting continuity of care, the proposed solution has the potential to create both practical healthcare value and long-term business value.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

The hardware budget for this research will mainly cover the basic equipment needed to collect caregiver voice data, develop the prototype, and test the system. This includes a laptop with sufficient processing power for model development, data processing, and system integration. In addition, a good-quality microphone will be required for recording caregiver voice logs clearly. If testing is carried out in a more realistic setting, a smartphone or tablet may also be used for voice input and system access.

These hardware items are necessary because the proposed system depends on voice capture, speech processing, NLP, reporting, and multimedia output. A suitable computer is required for model training, testing, and integration, while the microphone is important for collecting clear audio data. A mobile device is useful because caregivers often work in dynamic environments, so testing on portable devices can better reflect practical use.

7.2 Software Licenses

Most of the development can be done using open-source tools, which helps reduce software expenses. However, some costs may still arise if premium tools, specialized APIs, or commercial platforms are required. Software-related expenses may include licenses for development tools, database tools, 3D avatar platforms, or specialized Text-to-Speech services if free versions are not sufficient for research and demonstration purposes.

This expense is justified because some advanced features of the proposed system, especially avatar-based handover presentation and higher-quality TTS output, may require access to licensed tools or premium services. These tools can improve the quality, usability, and professionalism of the final prototype.

7.3 Cloud Services

Cloud-related expenses may include the cost of cloud storage, cloud computing, and model hosting or API usage if required during development and testing. Cloud services can be useful for storing collected voice data securely, running large AI models, and testing the scalability of the system. If the project uses Google Colab or similar environments for experimentation, there may also be costs related to premium compute access for faster processing and larger experiments.

This cost is justified because AI-based systems that involve speech processing, NLP, summarization, and multimedia generation may require more computational resources than a personal device can provide. Cloud services can support faster experimentation, efficient data handling, and easier prototype testing.

7.4 Human Effort Cost

Human effort cost includes the time and expertise required for system design, data collection, data annotation, model development, testing, and evaluation. This may involve the effort of the researchers, supervisors, technical assistants, domain experts such as caregivers or healthcare professionals who provide guidance and validation. If annotations are needed for ADL categories, disfluencies, or handover summaries, additional effort may also be required from human annotators.

This expense is justified because the development of the proposed system is not limited to technical implementation only. It also requires domain understanding, dataset preparation, validation, and expert feedback. Human effort is one of the most important parts of the project because accurate caregiver-oriented system design depends on both technical knowledge and healthcare-related input.

7.5 Estimated Budget Summary

A simple estimated budget plan for the proposed research is as follows:

- Hardware costs - computer, microphone, and mobile device support
- Software licenses - avatar tools, premium TTS tools, and other specialized platforms
- Cloud services - compute resources, storage, and model experimentation support
- Human effort cost - research, annotation, testing, and expert consultation

The exact values will depend on the final implementation choices, available institutional resources, and whether some tools are already provided by the university or research lab. If open-source tools and existing hardware are used, the total budget can be reduced significantly.

Overall Justification

The proposed budget is justified because each expense directly supports one or more important parts of the research. Hardware is required for data collection and development, software tools support advanced AI and multimedia features, cloud services provide computational support, and human effort is essential for system building and validation. Since the research aims to produce a working intelligent documentation prototype with speech, NLP, reporting, Text to Speech, and 3D avatar handover features, the budget is necessary to ensure that the final system is functional, reliable, and suitable for evaluation.

8. WORK BREAKDOWN STRUCTURE

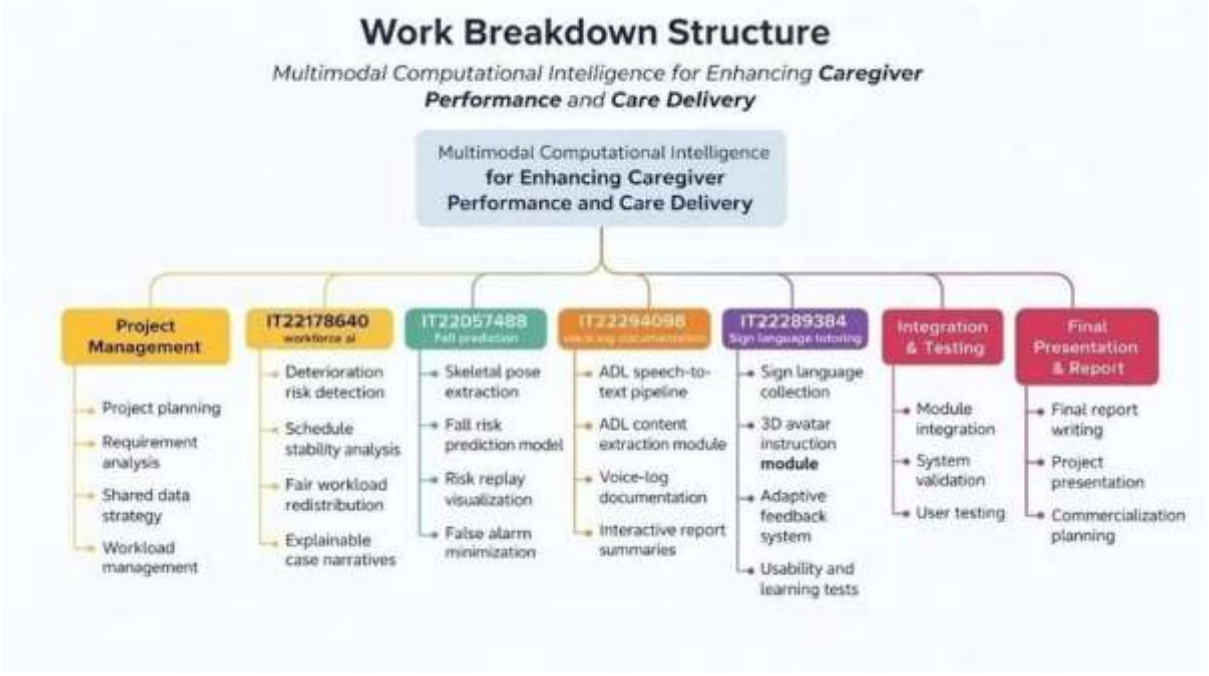


Figure 3 Work Breakdown Structure

9. GANTT CHART

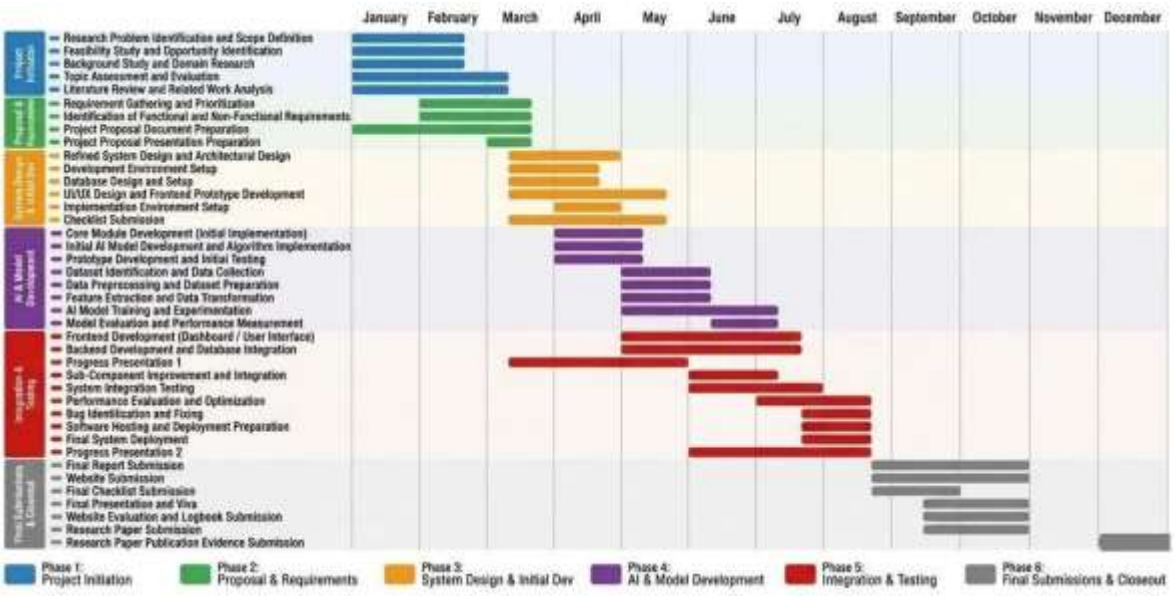


Figure 4 Gantt Chart

10.AI USAGE DISCLOSURE

AI tools were used in a limited and supportive capacity during the preparation of this research proposal. Tools such as ChatGPT were mainly used to improve language clarity, refine sentence structure, and support the organization and presentation of ideas. Their use was restricted to enhancing the readability and overall quality of the document, while ensuring that the main research concepts, problem definition, and proposed technical direction were independently developed by the author.

The development of all major aspects of this work, including the research problem, proposed solution, design choices, and methodological direction, was carried out independently. AI tools were not used to generate the full content of the proposal or to replace original thinking. Any AI-assisted suggestions were carefully reviewed, revised where necessary, and checked against academic and technical sources to ensure accuracy, relevance, and alignment with the research objectives. The author retains full responsibility for the originality, integrity, and content of this proposal.

I hereby declare that:

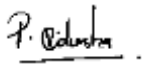
1. I have read and understood the AI Usage Policy for the Research Project (IT4010).
2. I acknowledge that AI tools are to be used only as support tools, not as substitutes for my own understanding and work.
3. I agree to use AI tools responsibly, ethically, and transparently.
4. I will clearly disclose any use of AI tools in my submissions.
5. I understand that I bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. I confirm that I will be able to explain, justify, and defend all aspects of my work during evaluations.
7. I understand that failure to comply with this policy may result in academic penalties.

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- Project Title: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs
- Group ID: R26-DS-010

Signature

- Student Signature:



- Date: 24/3/26

11.REFERENCES LIST

[1] Hirschmanner et al., "Prototype of a Care Documentation Support System Using Audio Recordings of Care Actions and Large Language Models," 2024. [Online]. Available: http://hirschmanner.com/publication/hllmi-2024/hllmi2024.pdf

[2] "Automated Generation of Discharge Summaries: Leveraging Large Language Models with Clinical Data," *Nature Scientific Reports*, 2025. [Online]. Available: https://www.nature.com/articles/s41598-025-01618-7

[3] "Natural Language Processing Systems for Extracting Information from Electronic Health Records about Activities of Daily Living," *PubMed*, 2024. [Online]. Available: https://pubmed.ncbi.nlm.nih.gov/38798774/

[4] "Context-Aware Medical Systems within Healthcare Environments: A Systematic Scoping Review," *MDPI International Journal of Environmental Research and Public Health*, vol. 20, no. 14, 2023. [Online]. Available: https://www.mdpi.com/16604601/20/14/6399

[5] "Privacy-Preserving On-Device Continual Adaptation of ASR for Clinical Telephony," *arXiv*, 2025. [Online]. Available: https://arxiv.org/html/2512.16401v4

[6] "Domain-Specific Adaptation for ASR through Text-Only Fine-Tuning," *ACL Anthology*, 2025. [Online]. Available: https://aclanthology.org/2025.mmloso-1.7.pdf

[7] "Disfluency Detection for Spoken Learner English," *ISCA Archive (SLATE)*, 2019. [Online]. Available: https://www.isca-archive.org/slate_2019/lu19_slate.pdf

[8] "Streaming Joint Speech Recognition and Disfluency Detection," *arXiv*, 2022. [Online]. Available: https://arxiv.org/pdf/2211.08726

[9] "Offline Speech Translation Systems: A Review of Opportunities, Technologies, and Future Prospects," *IJFMR*, 2025. [Online]. Available: https://www.ijfmr.com/papers/2025/6/62449.pdf

[10] "Intelligent Document Creation for a New Way of Working: Clinical Documentation Workflow with AI-Powered Speech Recognition," *Philips*, [Online]. Available: https://www.dictation.philips.com/fileadmin/landingpage_images/philipsdictation_intelligent-document-creation-for-a-new-way-of-working_medical_ebo_en.pdf

[11] "Improving the Robustness and Clinical Applicability of Automatic Respiratory Sound Classification Using Deep Learning-Based Audio Enhancement," *PMC*, 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC11950698/

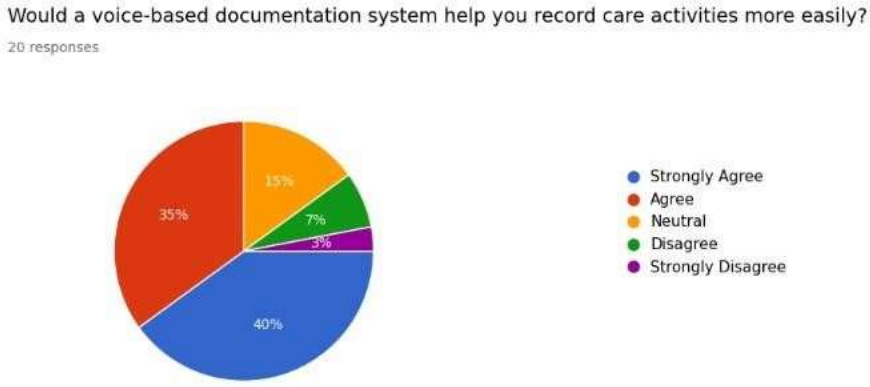
[12] "Fillers in Spoken Language Understanding: Computational and Psycholinguistic Perspectives," *ACL Anthology*, 2022. [Online]. Available: https://aclanthology.org/2022.tal-3.3.pdf

12.APPENDICES

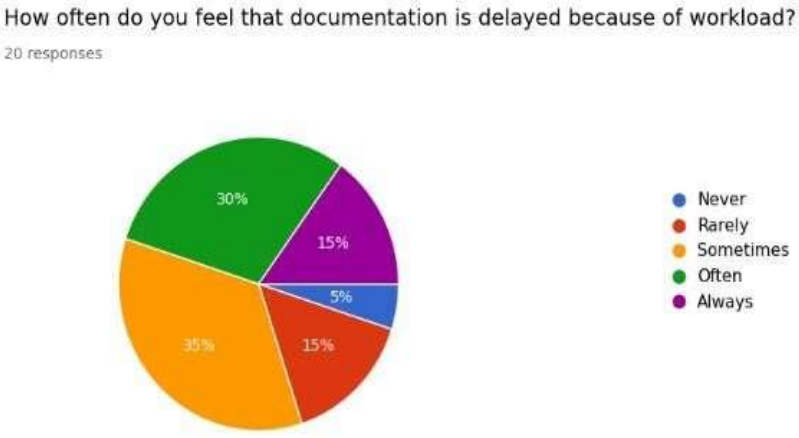
This appendix presents sample survey questions and response visualizations used to understand caregiver workflows, documentation challenges, and the need for a voice-based ADL documentation system.



Appendix 1 Difficulties Faced by Caregivers When Recording Daily Patient Care Activities



Appendix 2 Caregivers' Views on the Usefulness of a Voice-Based Documentation System



Appendix 3 Frequency of Documentation Delays Due to Workload